Clustering Algorithms

A Comparative Study

Mall Customers Dataset | 11 Algorithms

11	200	2	5
Algorithms	Customers	Features	Clusters

What We'll Cover

11 scikit-learn clustering algorithms applied to Mall Customers data

01	K-Means	Centroid-based, uses elbow method
02	Agglomerative	Hierarchical, bottom-up merging
03	Affinity Propagation	Message-passing, auto clusters
04	BIRCH	Tree-based, memory efficient
05	Bisecting K-Means	Divisive variant of K-Means
06	DBSCAN	Density-based, handles noise
07	Feature Agglomeration	Feature-side clustering
08	Mean Shift	Mode-seeking, bandwidth-based
09	Spectral Clustering	Graph-based, manifold aware
10	OPTICS	Density, variable eps
11	Gaussian Mixture	Probabilistic, soft assignments

Dataset Overview

Mall Customers · 200 rows · 5 columns · Features: Annual Income & Spending Score

CustomerID

Unique ID for each customer

Genre

Male / Female

Age

Customer age in years

Annual Income

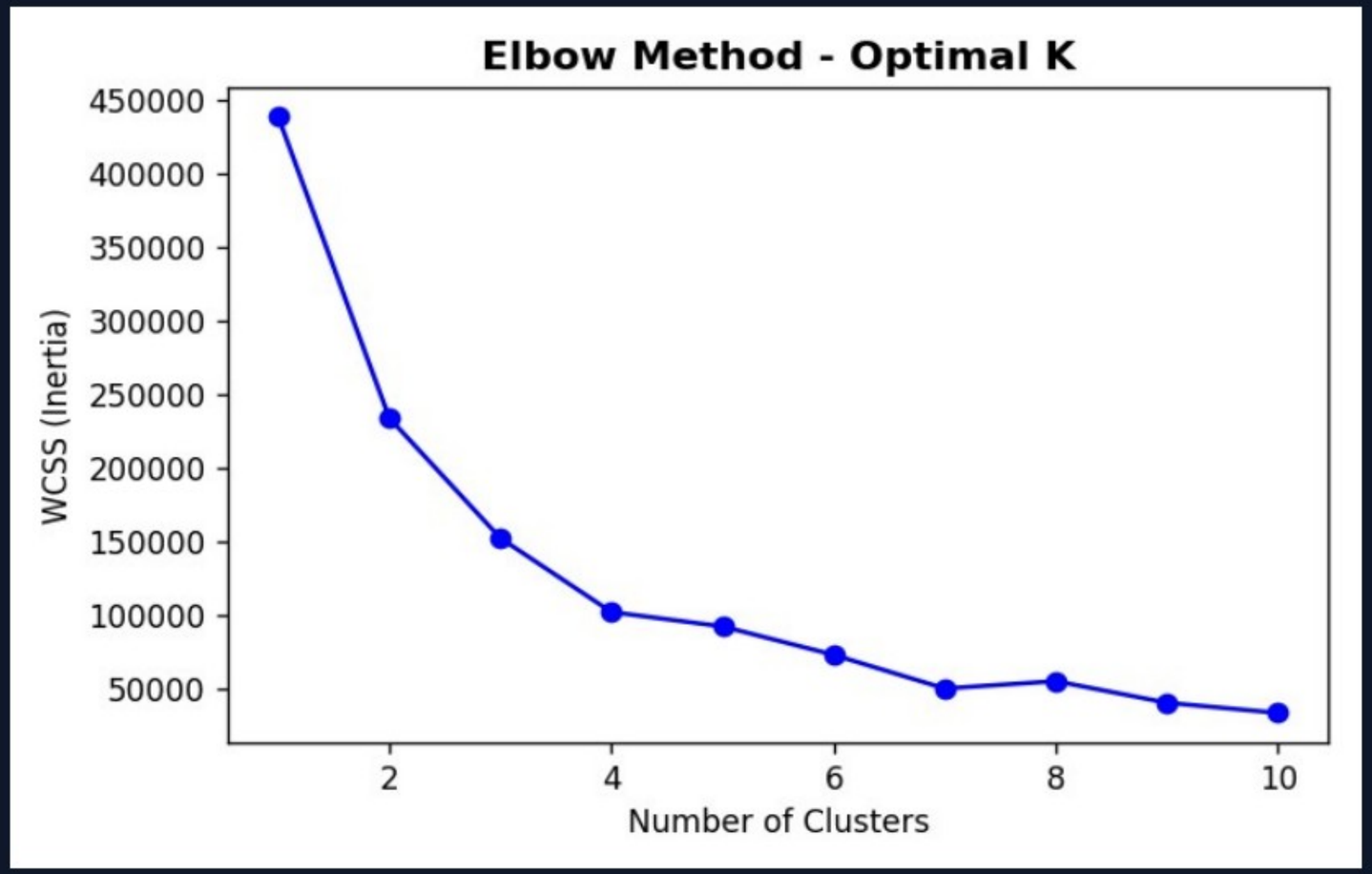
In thousands of dollars

Spending Score

Score 1–100 assigned by mall

Clustering targets the 2 key behavioral features: Annual Income & Spending Score

Elbow Method — Optimal Cluster Count



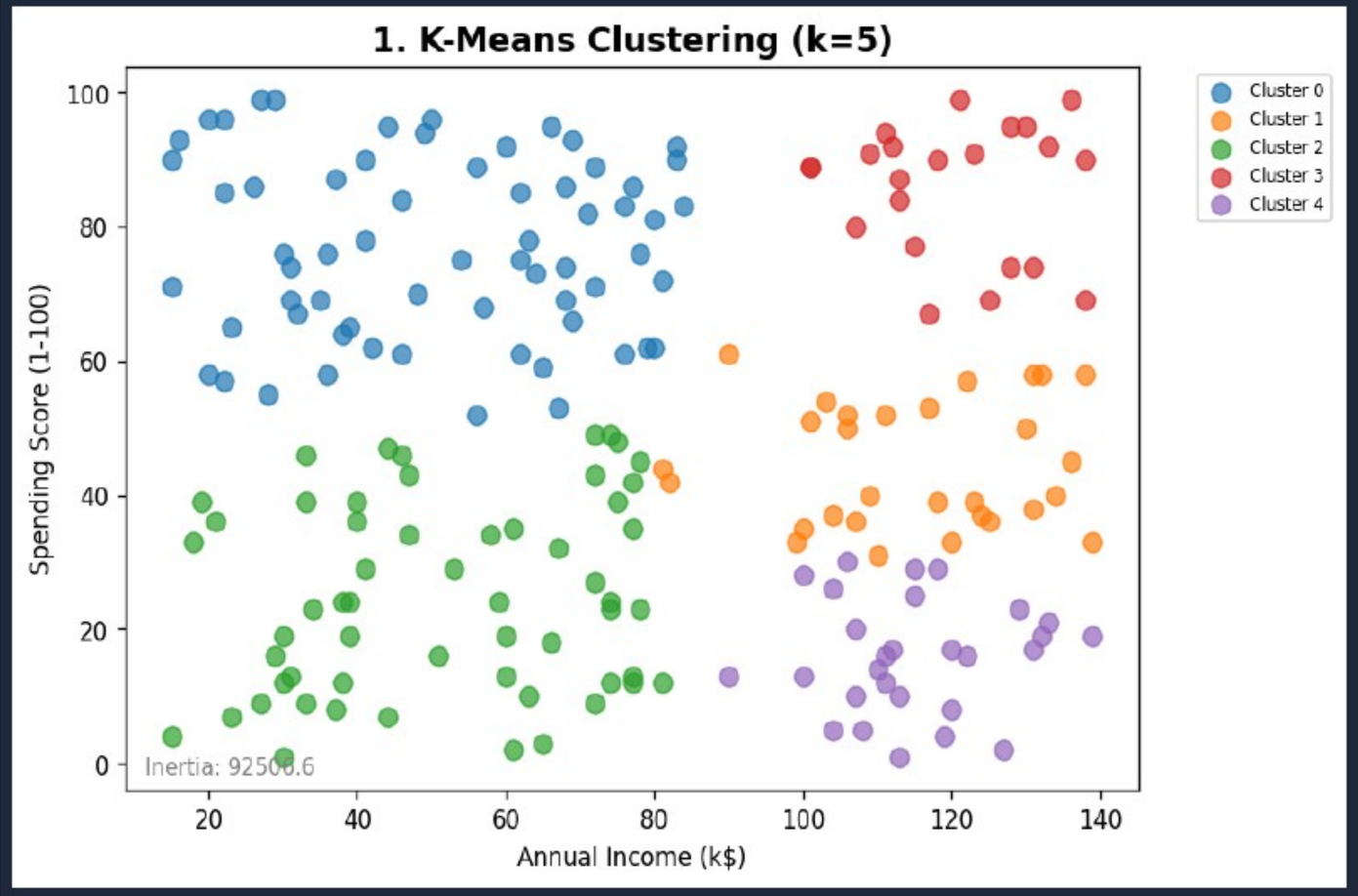
Key Observations

- WCSS drops sharply from k=1 to k=5
- Elbow point clearly at k = 5
- Used as reference by K-Means, BisectingKMeans, Spectral & GMM

01

K-Means Clustering

Centroid-based · n_clusters=5, init=k-means++



✓ Pros

- Fast & scalable
- Globally consistent clusters
- Interpretable centroids

✗ Cons

- Must specify k
- Sensitive to outliers
- Assumes spherical clusters

Result

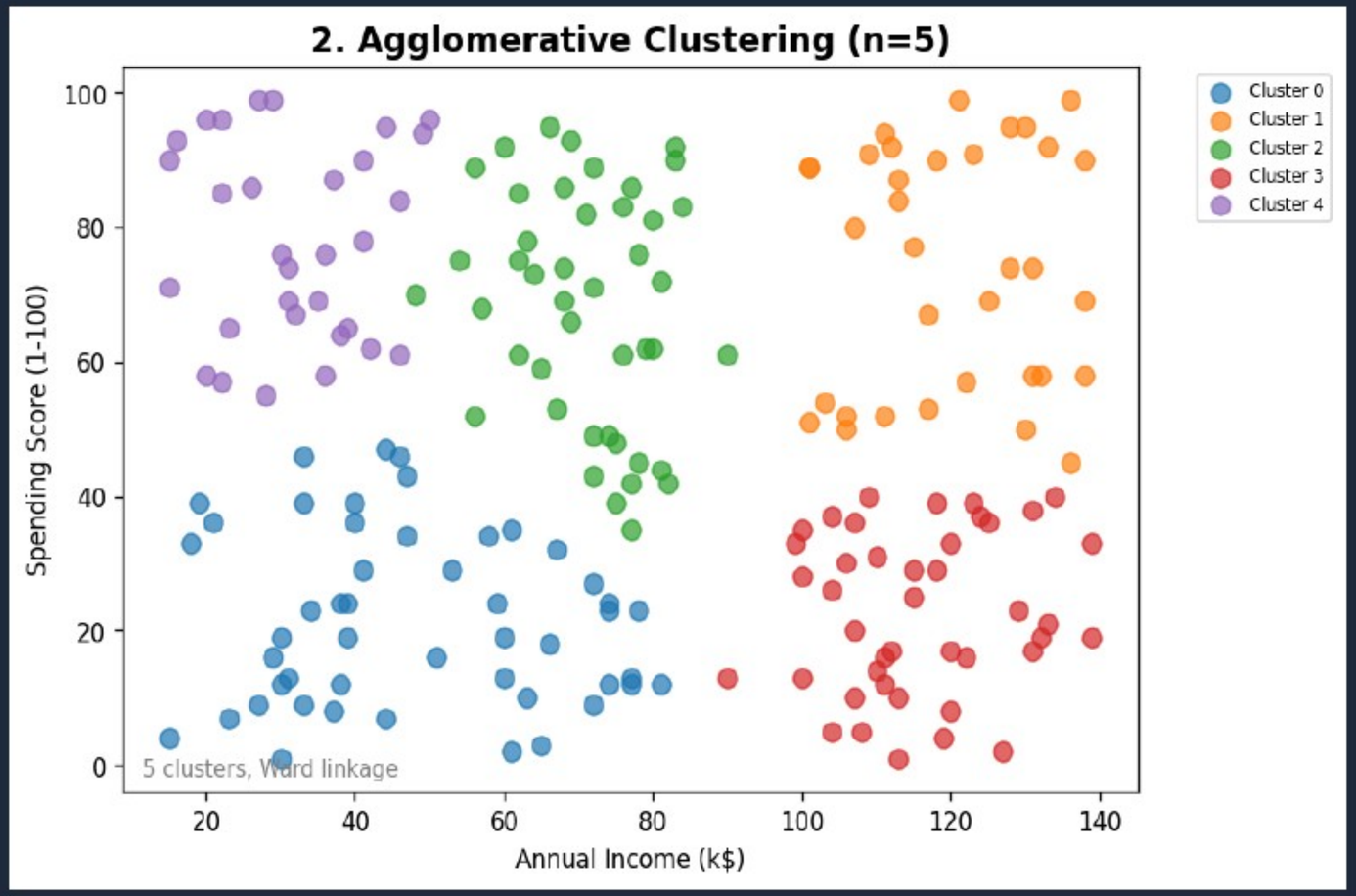
5 well-separated customer segments identified. Clear income–score groupings.

★★★★★ **Excellent**

02

Agglomerative Clustering

Hierarchical · n_clusters=5, linkage=Ward



✓ Pros

- No need for random init
- Dendrogram gives insights
- Deterministic

✗ Cons

- $O(n^2)$ memory
- Must specify k upfront
- Slower on large data

Result

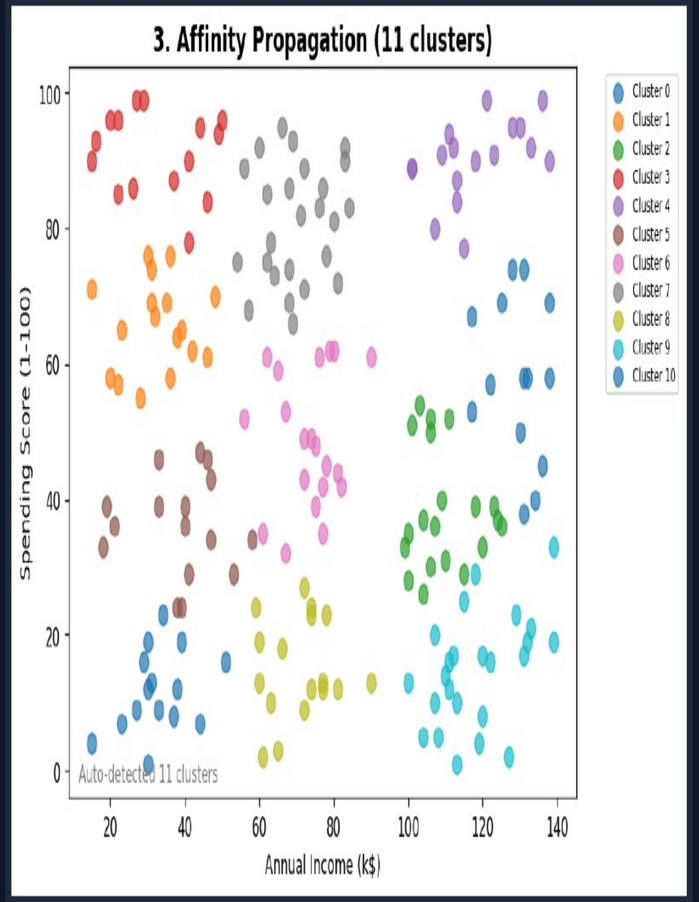
Clean 5-cluster split similar to K-Means; dendrogram confirms k=5.

★★★★☆ Very Good

03

Affinity Propagation

Message-Passing · random_state=15



✓ Pros

- Auto-detects cluster count
- No random initialization
- Exemplar-based

✗ Cons

- $O(n^2)$ complexity
- Many clusters possible
- Sensitive to damping

Result

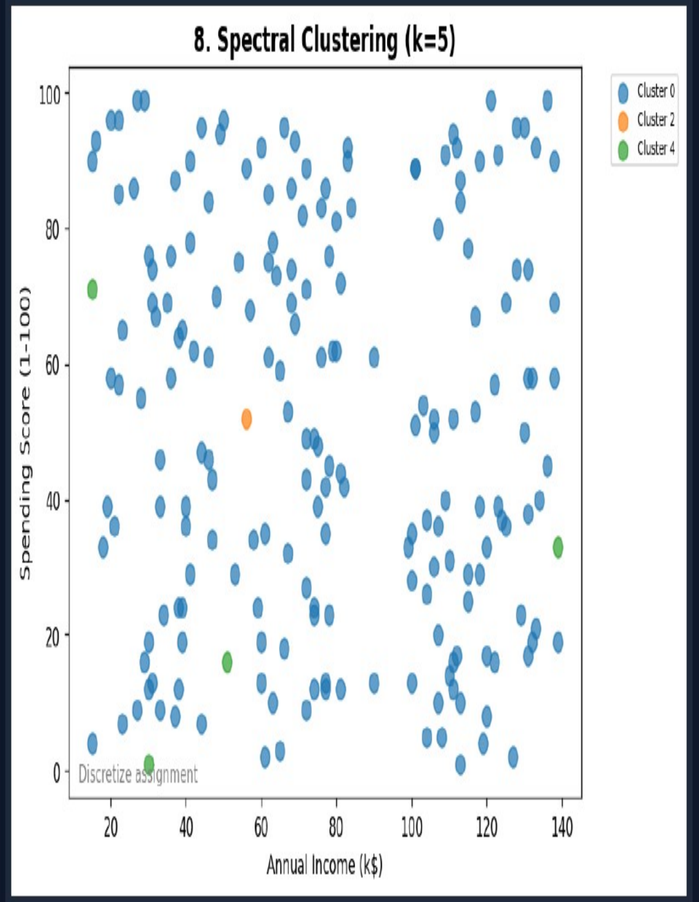
Many small clusters detected — may over-segment the mall data.

★★★☆☆ Moderate

09

Spectral Clustering

Graph-Based · n_clusters=5, assign_labels=discretize



✓ Pros

- Handles non-convex shapes
- Powerful for complex manifolds

✗ Cons

- Expensive $O(n^3)$
- Needs k upfront
- Sensitive to graph params

Result

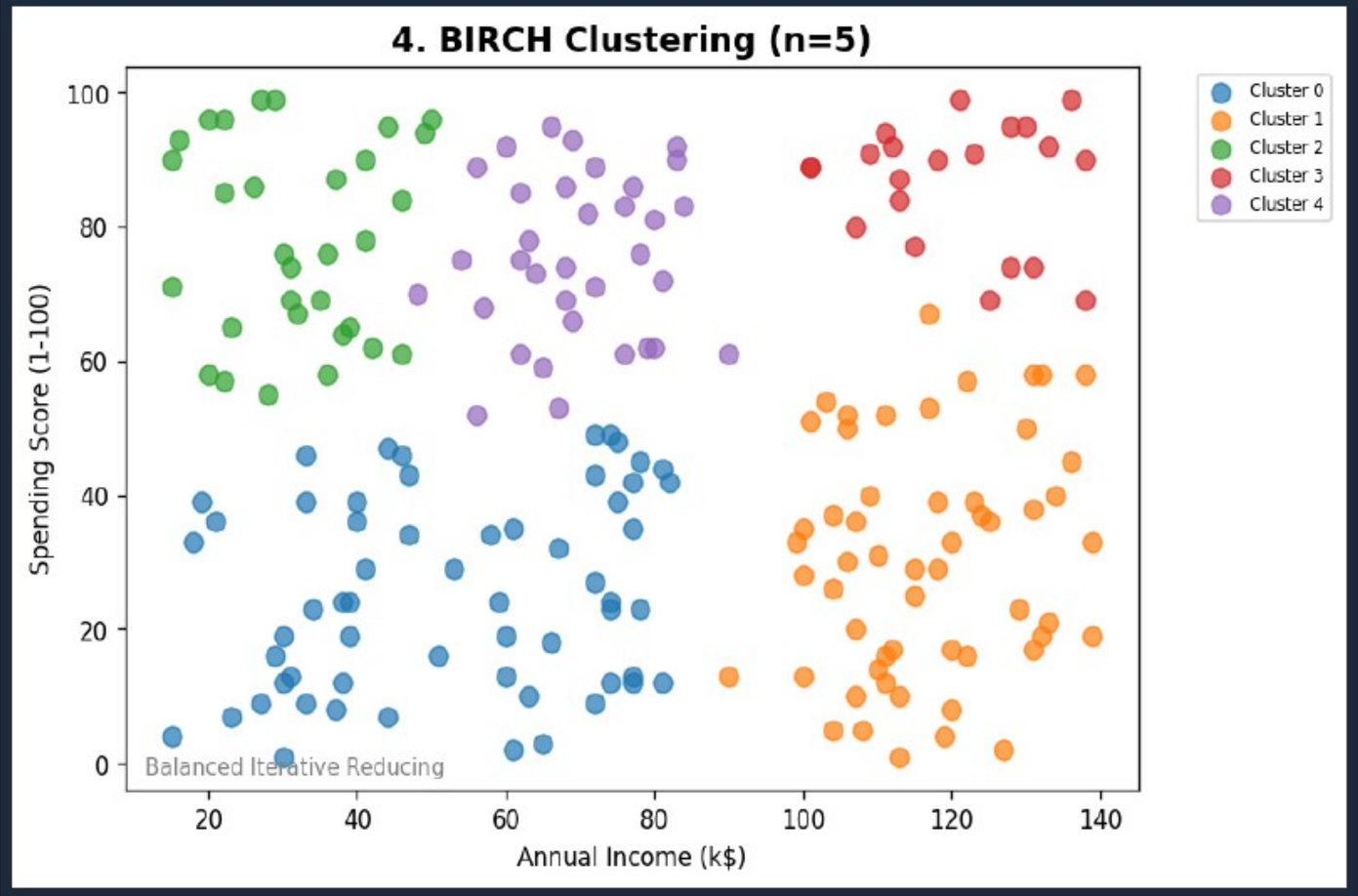
5 clusters formed; convergence warning noted — data may be too uniform.

★★★☆☆ Moderate

04

BIRCH

Tree-Based · n_clusters=5



✓ Pros

- Memory efficient
- Fast on large datasets
- Good for streams

✗ Cons

- Less effective in high dimensions
- Sensitive to threshold

Result

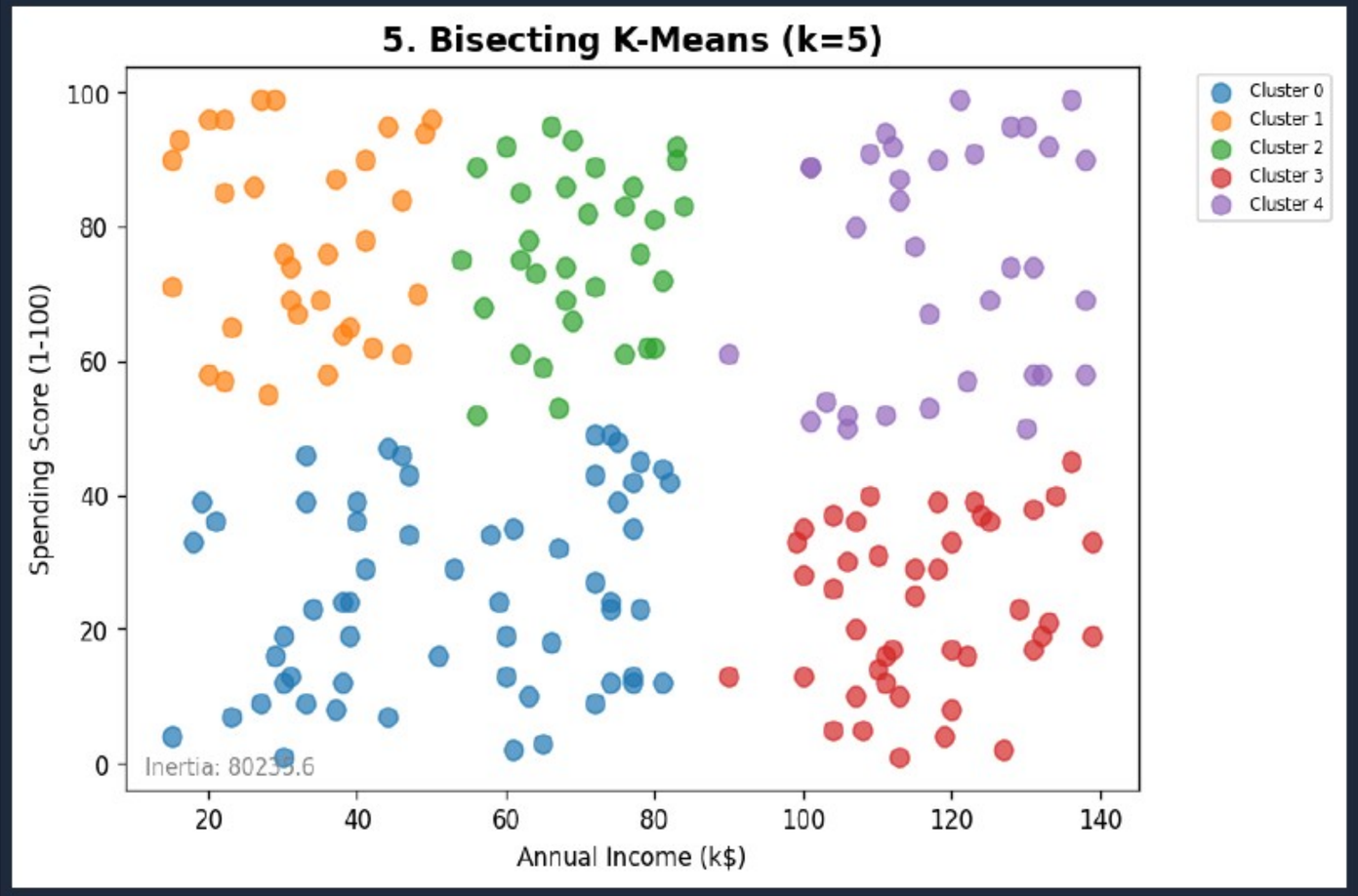
Produces 5 clusters comparable to K-Means; efficient performance.

★★★★☆ Very Good

05

Bisecting K-Means

Divisive · n_clusters=5, random_state=42



✓ Pros

- More stable than standard K-Means
- Hierarchical structure
- Scalable

✗ Cons

- Bisection decisions are final
- Still needs k specified

Result

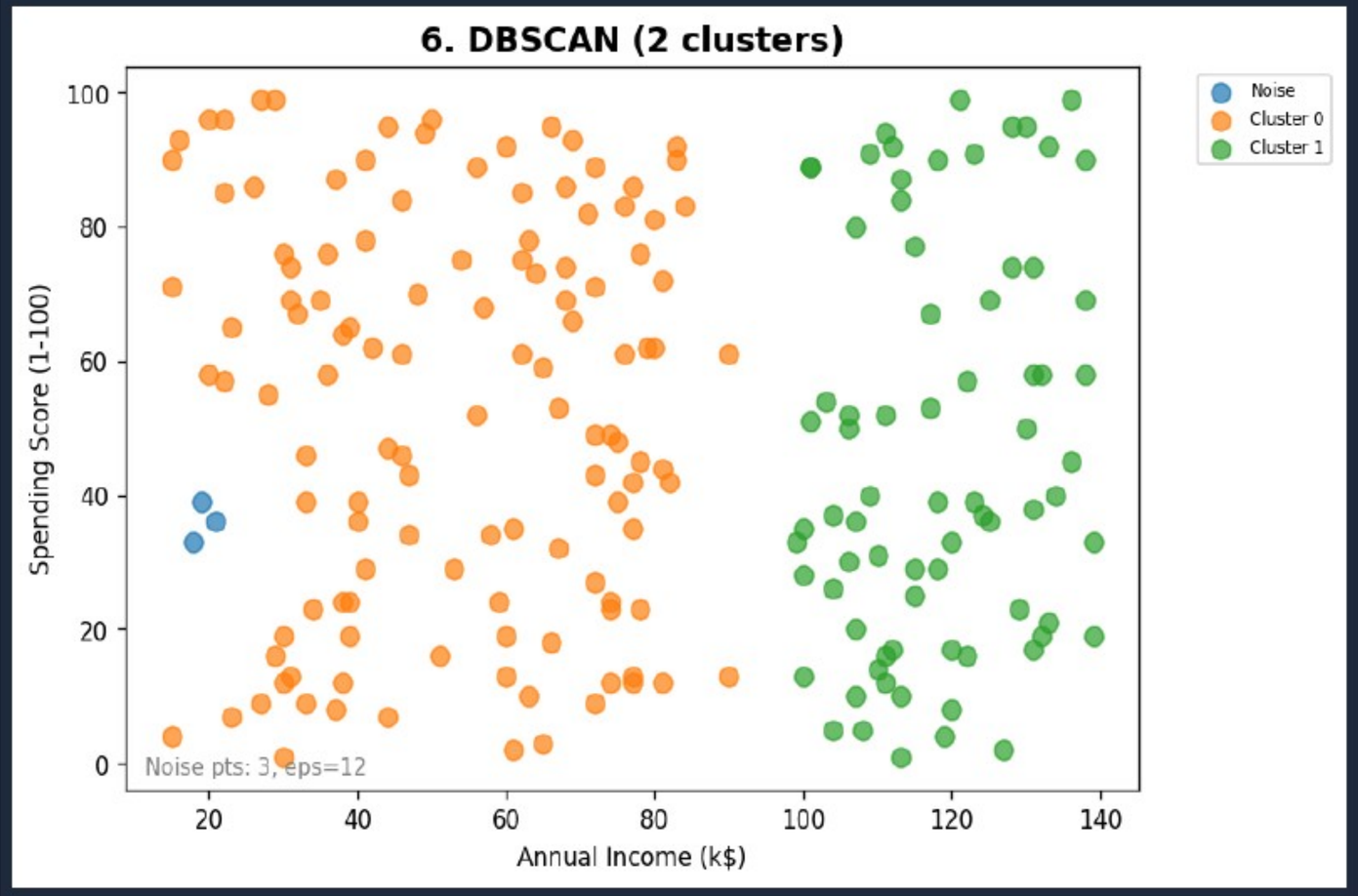
Results closely match K-Means with slight boundary differences.

★★★★☆ Very Good

06

DBSCAN

Density-Based · eps=12, min_samples=4



✓ Pros

- No need to specify k
- Finds arbitrary shapes
- Handles noise/outliers

✗ Cons

- eps tuning is critical
- Struggles with varying density
- Not great for uniform data

Result

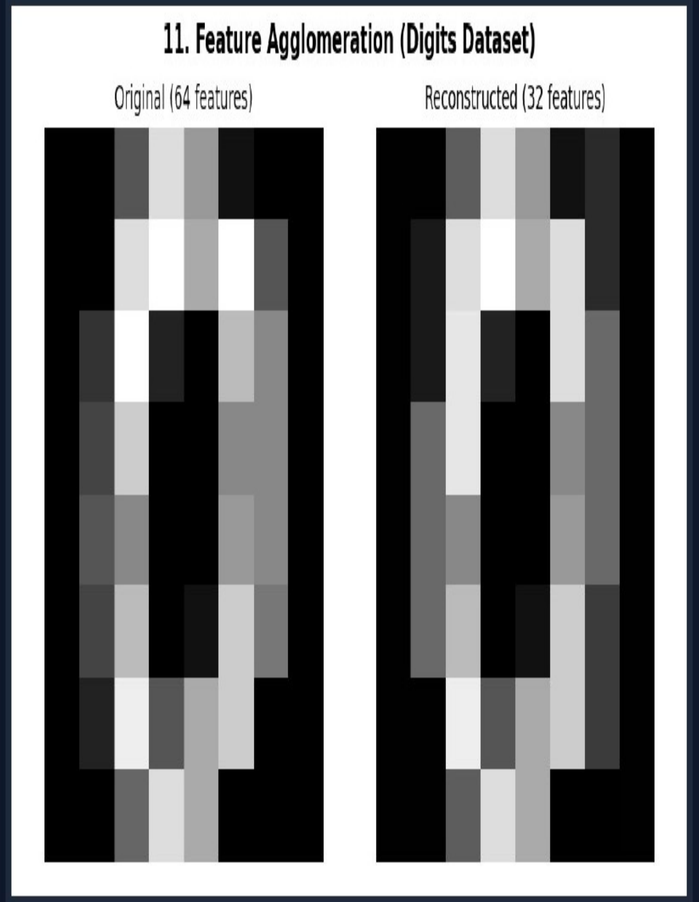
Handles noise well; cluster count depends on eps — sensitive to tuning.

★★★☆☆ **Context-Dependent**

07

Feature Agglomeration

Feature-Side Clustering · n_clusters=32 (features)



✓ Pros

- Reduces feature dimensionality
- Good for high-dim data
- Unsupervised compression

✗ Cons

- Not sample clustering
- Lossy compression
- Loses some fine details

Result

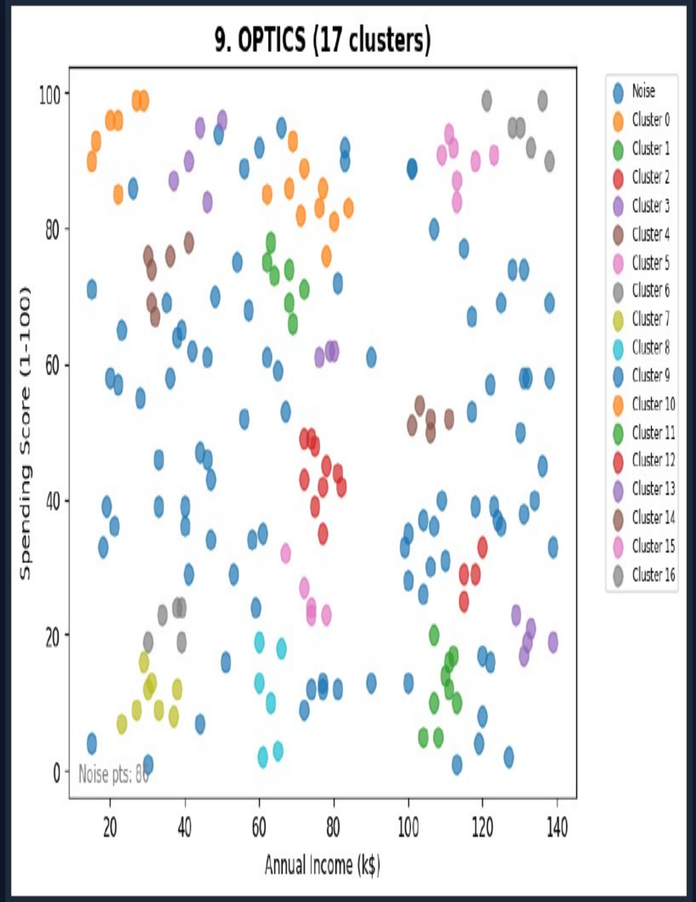
Reduces 64 digit features to 32 — good reconstruction fidelity shown.

★★★☆☆ Different Purpose

10

OPTICS

Density-Based (Enhanced) · min_samples=4



✓ Pros

- Variable density handling
- No need to specify eps
- Hierarchical output

✗ Cons

- Complex output interpretation
- Slow on large datasets
- Hard to tune

Result

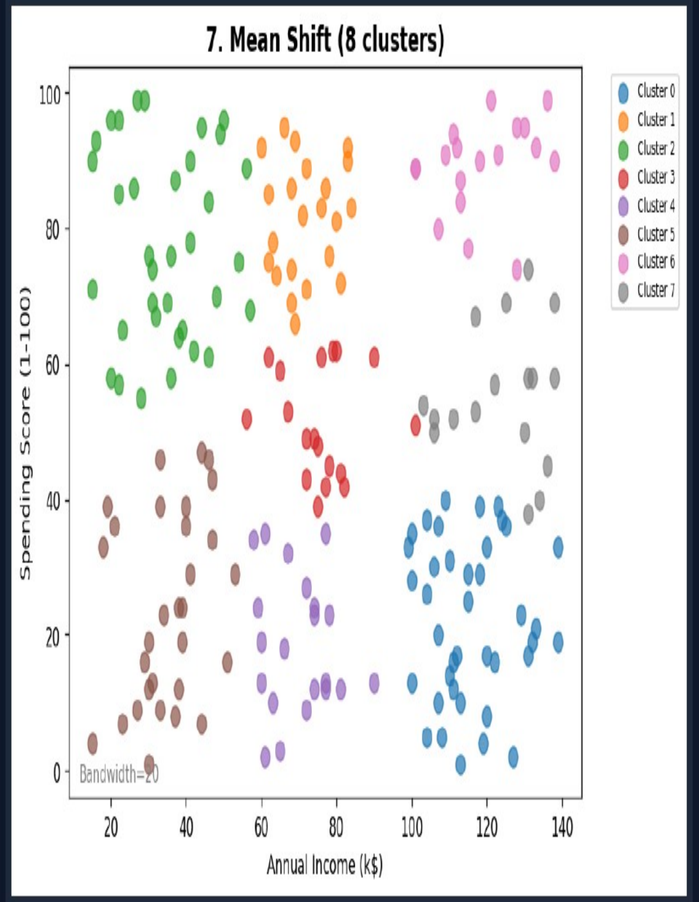
Detects density-connected clusters but may label many points as noise.

★★★☆☆ Moderate

08

Mean Shift

Mode-Seeking · bandwidth=20



✓ Pros

- Auto-detects clusters
- No assumed shape
- Robust to outliers

✗ Cons

- Bandwidth must be set
- Slow $O(n^2)$
- Merges nearby clusters

Result

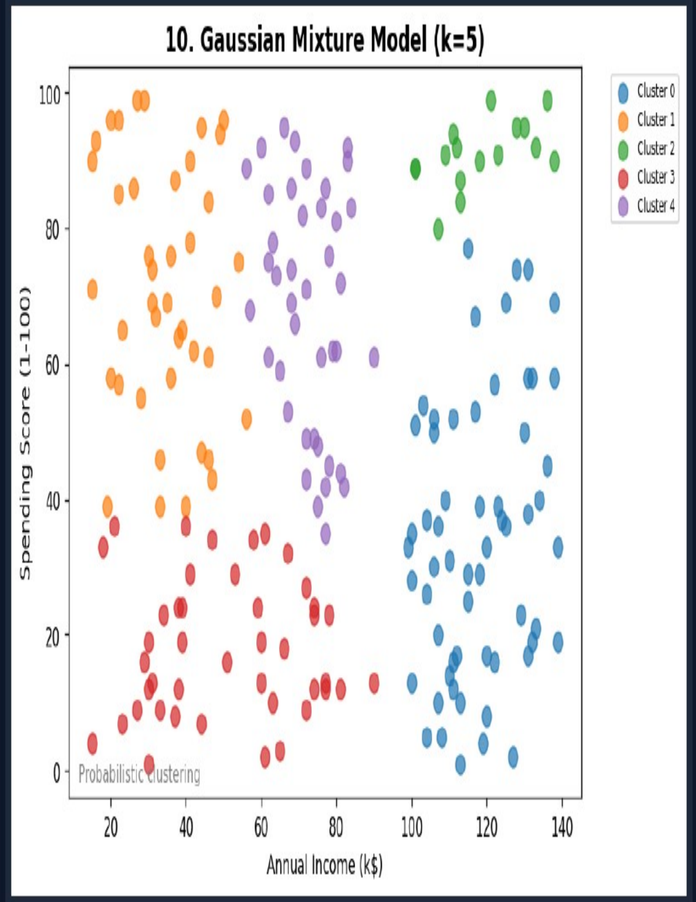
Good separation with bandwidth=20; compact well-centered clusters.

★★★★☆ Good

11

Gaussian Mixture Model

Probabilistic · n_components=5, random_state=0



✓ Pros

- Soft cluster assignments
- Handles elliptical clusters
- EM-based, principled

✗ Cons

- Needs k specified
- Can overfit with many components
- Slower than K-Means

Result

Smooth probabilistic boundaries; slightly different from hard clustering.

★★★★☆ Very Good

Algorithm Comparison & Verdict

Algorithm	Clusters	Noise	Speed	Interpretability	Rating
K-Means	5 (set)	None	Fast	High	★★★★★
Agglomerative	5 (set)	None	Med	High	★★★★☆
Affinity Prop	Auto	None	Slow	Med	★★★☆☆
BIRCH	5 (set)	None	Fast	Med	★★★★☆
Bisecting K-M	5 (set)	None	Fast	High	★★★★☆
DBSCAN	Auto	Yes	Med	Med	★★★☆☆
Feature Agglo	N/A	N/A	Fast	Low	★★★☆☆
Mean Shift	Auto	None	Slow	Med	★★★★☆
Spectral	5 (set)	None	Slow	Low	★★★☆☆

🏆 **BEST for Mall Data: K-Means (k=5) — Fast, interpretable, clean 5-segment split confirmed by elbow method**